

Positional Semantics for Retrieved Native-KV Memory: RoPE, Fragmentation, and Bounded Relocation in Progressive Retrieval Attention

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Abstract

Chunked and retrieved key/value (K/V) memory creates a positional problem that ordinary prefix caching largely avoids: a token may be encoded in a short local block even though it belongs at a later position in its source. Resetting every block to zero changes the state that is cached and the attention relation observed when that state is retrieved. We study this problem in Progressive Retrieval Attention (PRA), which routes through compact gists but materializes selected full-detail native K/V.

The general repair is a source-relative logical offset. Across learned absolute, sinusoidal, and rotary position embedding (RoPE) models, offsets remove layer-0 reset error exactly and reduce final-layer K error in all 30 paired comparisons over two capacities and five seeds. RoPE contributes something more specific: pre- and post-RoPE keys are exactly equivalent at matched effective positions, while storing pre-RoPE keys permits retrieval-time rebinding. In the measured unfused CUDA path, that flexibility raises warm latency from 0.179 to 0.649 ms.

The main discovery is a separation, not a universal quality gain. On WikiText-2, offsets improve final-layer representation in 30/30 pairs but next-token loss in only 5/30. On controlled HotpotQA- and QASPER-derived probes, routed loss improves in 35/60 pairs, with the direction depending on position mechanism. Overlap reduces contextual fragmentation at additional encoding cost, yet task effects remain mixed. A 192-token logical prompt also executes with every measured native operation at or below 24 tokens under a 32-token limit. Positional continuity is therefore necessary for faithful chunked K/V, but coordinate correctness, representation fidelity, attention geometry, and task usefulness must be evaluated separately.

1 Introduction

A key retrieved from external memory carries more than content. Its hidden state reflects the causal context present when it was encoded, and its use in attention reflects a position. If a long source is divided into bounded blocks and every block starts again at position zero, later tokens silently inherit the coordinates of earlier ones. The cache is then internally consistent as a tensor but wrong as a representation of the source.

This problem is broader than RoPE. Learned absolute embeddings and sinusoidal embeddings also change when a block is reset. RoPE is special for a different reason: because position is an explicit rotation of queries and keys, it provides an algebra for preserving or changing a cached key's binding. The offset requirement is general; the rebinding operation is rotary.

We study both in PRA, a sparse native-KV memory mechanism. A long source or displaced prompt head is encoded in bounded blocks. Compact gists route a later query to selected chunks,

but attention consumes the chunks’ original token-level K/V. The positional semantics of those stored keys must therefore be specified rather than left as an implementation accident.

The paper is organized around four increasingly demanding questions:

1. **Coordinate correctness:** were source-relative positions preserved?
2. **Representation fidelity:** does chunked K/V resemble the state produced with the intended coordinates and context?
3. **Attention-geometry fidelity:** does the positional treatment reproduce the intended query–key relation?
4. **Task usefulness:** does the corrected state improve language-model loss, evidence selection, or answers?

These properties form a dependency, not an equivalence. Correct coordinates can improve a stored representation without reconstructing missing causal context. Faithful K/V can still be selected poorly. Even annotated evidence can be an inferior context composition for a particular model. The headline result is consequently:

Geometric correctness and functional usefulness are not the same thing.

Our contributions are outcome-oriented:

- a source-relative position contract for independently encoded K/V;
- evidence that logical offsets repair reset fragmentation for learned absolute, sinusoidal, and RoPE models, not only for rotary models;
- an exact pre-/post-RoPE equivalence test and a measured deferred-rotation tradeoff;
- a separation between coordinate repair and overlap-based contextualization;
- a cross-domain demonstration that representation improvements need not yield monotonic task improvements;
- bounded implicit-head execution in which logical prompt length exceeds every native model operation; and
- controls showing that evidence identity, routing, and useful memory composition remain separate from positional correctness.

2 The Problem in One Example

Suppose a source span occupies positions 100 through 131. Independent chunk encoding often does this:

View	First token	Last token
Source/physical position	100	131
Wrong chunk-local reset	0	31
Correct logical/effective position	$100 + 0$	$100 + 31$

The correction is simply

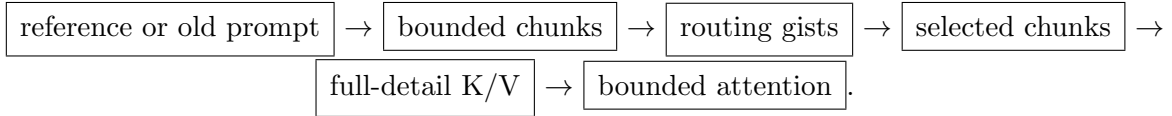
$$p_{\text{effective}} = p_{\text{chunk,start}} + p_{\text{inside}}, \quad (1)$$

where $p_{\text{chunk,start}}$ is stored once for a contiguous chunk and $p_{\text{inside}} \in [0, T - 1]$ is its local index. The formula restores the coordinate the token owns inside its source.

The word *source-relative* is important. Positions 100–131 are meaningful within one document, reference, or displaced prompt head. Two unrelated documents do not acquire a meaningful global ordering merely because both are registered in one request. Source identity and logical offset are therefore cache provenance. A continuously advancing `#_head`, by contrast, represents one prompt history and retains one coordinate system as direct tokens roll into memory.

3 PRA and the Four Correctness Levels

PRA is self-contained in the following path:



The gist is routing metadata, not a compressed substitute for the selected token state. Once a chunk is selected, its full token-level native K/V joins direct-context K/V under the same attention softmax. Position errors introduced while building that K/V survive routing and materialization; the router cannot repair them afterward.

Three lengths remain distinct. L_{logical} is the addressable source span, $L_{\text{model-op}}$ is the maximum K/V participating in one native operation, and L_{position} is the supported coordinate range. PRA can make $L_{\text{logical}} > L_{\text{model-op}}$ by caching and selecting bounded regions, but it does not by itself guarantee that a model has learned useful behavior at every L_{position} .

Table 1 summarizes the empirical answers before the details.

Table 1: Headline results. Each row answers a different correctness question.

Question	Verified result	Meaning
Do offsets repair reset coordinates?	Layer-0 K RMSE becomes zero in 30/30 paired comparisons	Source-relative continuity is a general requirement
Do deeper representations improve?	Final-layer K RMSE falls in 30/30 pairs	Coordinate repair survives two capacities and three mechanisms
Does overlap improve representation?	Yes in the verified WikiText scope; 29/30 loss pairs also improve modestly	Contextualization is separate and costs extra encoding
Are pre/post RoPE keys equivalent?	Zero K, logit, and output error at matched effective positions	Storage state does not change the intended geometry
Does better geometry guarantee task quality?	No: WikiText loss improves in 5/30; QA routed loss in 35/60	Functional utility is a separate empirical property
Can deferred rotation rebind keys?	Yes; warm latency rises from 0.179 to 0.649 ms	Flexibility has a measured systems cost
Can logical prompt exceed a native operation?	192 logical tokens, maximum measured operation 24 under a limit of 32	Bounded positional and transport mechanics

The failure modes align with the four levels:

Table 2: No single intervention guarantees all four kinds of correctness.

Intervention	What it repairs	What remains
Logical offsets	Reset-position fragmentation	Missing causal context, routing, composition
Encoding overlap	Some contextual fragmentation	Extra encoding cost; mixed task response
Matched RoPE binding	Intended relative attention geometry	Semantic retrieval and learned distance use
Deferred RoPE	Retrieval-time positional rebinding	Additional runtime and adaptation evidence
Oracle evidence	Selection uncertainty	Whether that evidence set is the best composition

4 What RoPE Uniquely Adds

4.1 Intuition Before Algebra

RoPE rotates each query and key according to its position. Attention sees the difference between those rotations, so translating both a query and key together preserves their relative geometry. Translating only the key changes its distance from a fixed query and is not an invariant operation.

This yields two storage choices:

Post-RoPE K	Pre-RoPE K
Position is bound when the cache is written. Retrieval is cheap, but the binding is fixed.	Position is applied when the cache is read. Retrieval-time rebinding is possible, with extra work.

4.2 The Minimal Algebra

For a head of width d_h , let R_p denote the RoPE rotation at position p . The rotated query and key are

$$\tilde{q}_i = R_i q_i, \quad \tilde{k}_j = R_j k_j. \quad (2)$$

Because $R_i^\top R_j = R_{j-i}$,

$$\tilde{q}_i^\top \tilde{k}_j = q_i^\top R_{j-i} k_j. \quad (3)$$

Equation 3 answers one question: why does common translation preserve the query-key relation? Adding c to both positions leaves $j - i$ unchanged. It does not say that $R_i q_i$ has the same relation to $R_{j+c} k_j$.

Pre- and post-RoPE storage are algebraically equivalent when they use the same effective position:

$$K_{\text{post}}(p) = R_p K_{\text{pre}}. \quad (4)$$

The four-cell model test confirms zero K, logit, and output RMSE, with top-token agreement 1.0, for matched offset positions across both tiers, all layers, and five seeds. Reset binding instead has mean K RMSE 0.648 and output RMSE 0.063. Deliberate rebinding by +512 changes the output, as it should.

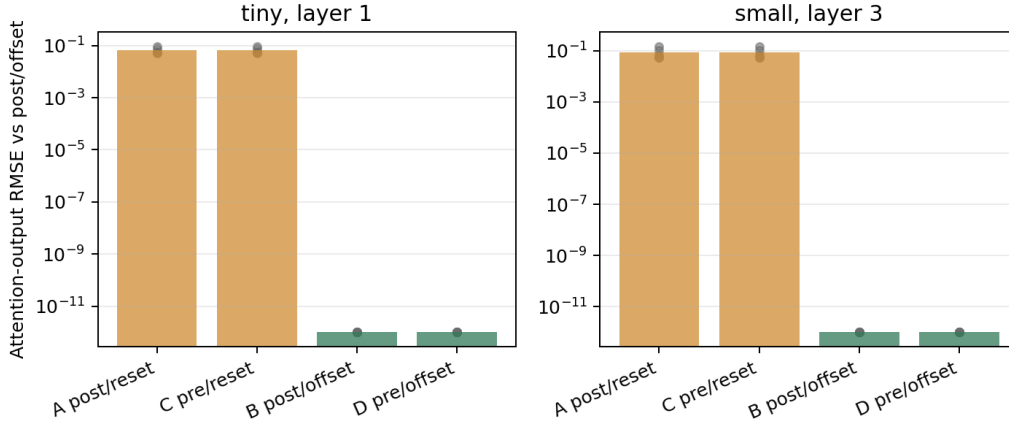


Figure 1: Matched pre- and post-RoPE storage reproduces the same attention geometry exactly. Reset coordinates and deliberate K-only rebinding change it.

The pure tensor test reaches the same conclusion. Translating Q and K together preserves the top-attended token, with maximum logit RMSE 5.79×10^{-5} through the measured shifts. Moving K alone by up to 8,192 positions changes fixed-query scores. This is the correct behavior, not an error in RoPE.

4.3 Deferred Rotation Has a Cost

At 160 selected small-tier tokens, storing post-RoPE K gives a 0.1791 ms warm path. Storing pre-RoPE K, reconstructing positions, and rotating at materialization gives 0.6487 ms. The position reconstruction itself is small; the unfused RoPE transform dominates. The nearly flat timing across token counts indicates launch-dominated behavior on the measured GTX 950M, so this is a prototype cost measurement rather than a production throughput prediction.

Both modes retain the same number of K/V elements. For 192 stored small-tier tokens, K/V uses 196,608 bytes. A contiguous chunk needs one 8-byte logical start for execution; a full 192-token int64 audit vector would use 1,536 bytes. Deferred rotation changes when phase is assigned, not the size of the native K/V payload.

5 General Offset Repair Beyond RoPE

The general experiment compares independently encoded 32-token blocks with a dense 128-token representation. It covers two model capacities, learned absolute, sinusoidal, and RoPE positions, and five matched seeds per condition. Table 3 reports layer-0 and final-layer K RMSE.

Table 3: Reset versus source-relative offsets. Every final-layer paired difference favors offsets; every offset layer-0 error is exactly zero.

Tier	Position	Reset L0	Offset L0	Reset final	Offset final	Improved
Tiny	Learned absolute	.6673	.0000	.6769	.1129	5/5
Tiny	Sinusoidal	.3640	.0000	.4722	.1436	5/5
Tiny	RoPE	.6463	.0000	.6745	.1373	5/5
Small	Learned absolute	.6626	.0000	.7193	.3066	5/5
Small	Sinusoidal	.3345	.0000	.5507	.2746	5/5
Small	RoPE	.5586	.0000	.7498	.4235	5/5

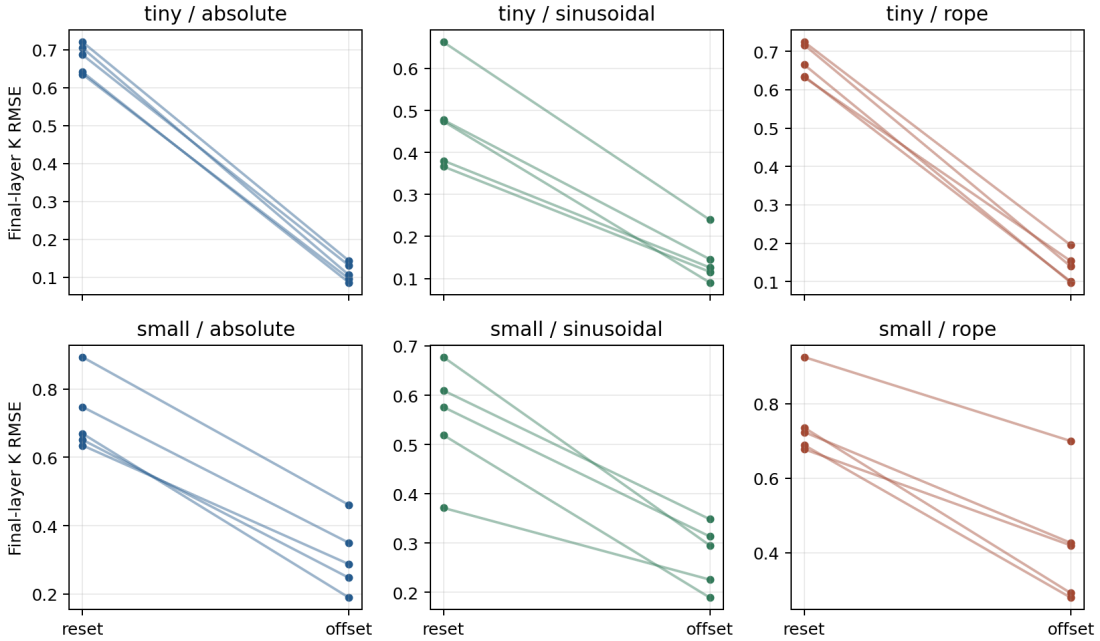


Figure 2: Offsets reduce final-layer K error in all 30 paired runs. The residual after exact layer-0 repair reveals a second problem: missing causal context across encoding blocks.

The sinusoidal result rules out a RoPE-only explanation. Logical offsets repair coordinate reset because they restore source position, regardless of whether position enters through an embedding table, a formula-defined additive embedding, or rotary Q/K geometry.

The residual final-layer error is equally informative. At layer zero, matched token content and effective positions suffice for exact equality. At later layers, a block lacks the causal left context available in dense encoding. This is contextual fragmentation, not failed coordinate repair.

6 Overlap Repairs a Different Problem

Encoding overlap gives each block additional native left context and commits only its new core. It does not change routing granularity and should not be described as a routing intervention. In the controlled representation test, 25% overlap lowers final-layer mean error in all six tier-mechanism conditions. At 50%, five of six improve; small learned absolute RMSE moves slightly from 0.3066 to 0.3103. The intervention is useful but not monotonic.

WikiText-2 gives a clear quality/cost point. At an 8:1 logical-to-native ratio, overlap lowers final-layer error in every condition and next-token loss in 29/30 paired runs relative to offset-only encoding. The loss reductions are modest, from 0.0007 to 0.0173 in the tiny tier and 0.0102 to 0.0148 in the small tier, while encoding processes 1.9375 times the unique source tokens. Overlap buys contextualization through recomputation.

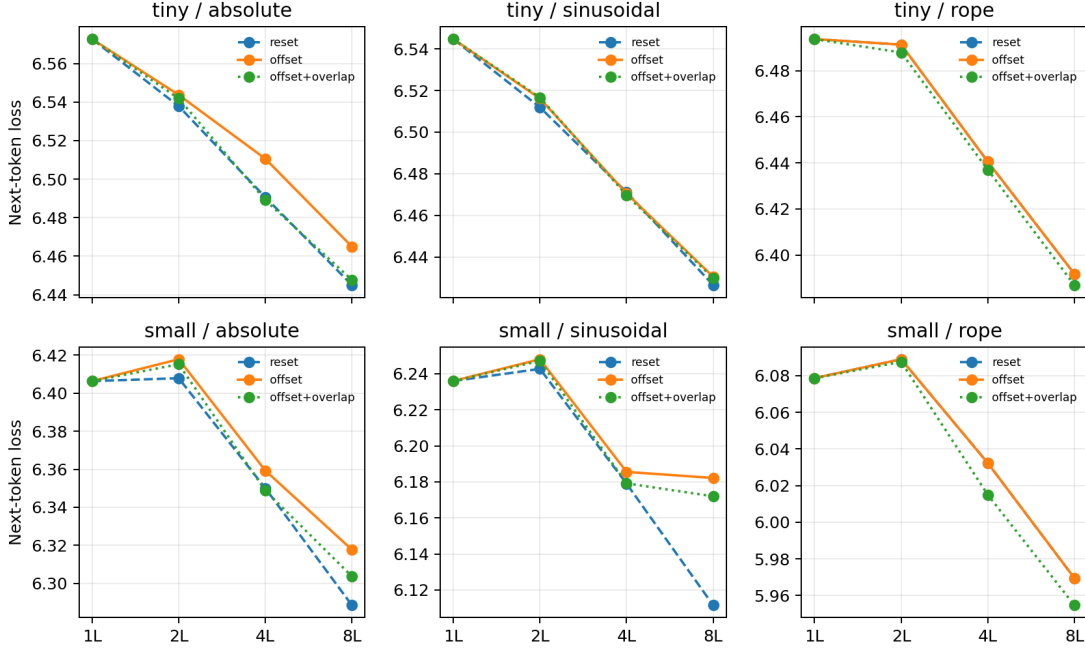


Figure 3: On WikiText-2, offsets sharply improve representation fidelity without a matching loss gain. Overlap gives smaller, more consistent loss changes at almost $1.94\times$ encoding cost at 8L.

Across the QA-derived probes, overlap effects are mixed and mechanism-dependent. That does not make overlap a failed coordinate method; it confirms that it repairs a different input to the task pipeline.

7 The Geometry–Utility Dissociation

The strongest scientific result appears when representation and task metrics are placed on the same axis. On 8L WikiText-2 windows, offsets reduce final-layer K RMSE in 30/30 paired runs but reduce next-token loss in only 5/30. Learned-absolute loss rises by 0.0200 in the tiny tier and 0.0292 in the small tier; sinusoidal loss rises by 0.0040 and 0.0707; RoPE loss is unchanged to the reported precision.

The controlled HotpotQA- and QASPER-derived answer-code probes likewise resist a universal story. Routed loss improves in 35/60 paired comparisons. Learned-absolute and sinusoidal means improve in both capacities and datasets, whereas RoPE means worsen in all four dataset–capacity cells. These probes use natural source text but simplified targets; they measure positional and retrieved-memory behavior, not unrestricted QA.

Every measured encoding and attention call contains at most 24 tokens; native-limit violations are zero in all six tier-mechanism conditions. Source-relative offsets lower mean answer loss in 27/30 paired seed comparisons. Overlap lowers it further in 23/30. The effect size remains mechanism-dependent, and the small RoPE mean stays high despite the mechanical correction.

Table 4: Continuous-head mean answer loss over four examples and five seeds. All rows use 192 logical positions, a 32-token operation limit, and a maximum observed operation of 24.

Tier	Position	Reset routed	Offset routed	Offset + overlap
Tiny	Learned absolute	2.3837	.7304	.6474
Tiny	Sinusoidal	2.9691	.9388	.8021
Tiny	RoPE	2.5910	1.5547	.8763
Small	Learned absolute	3.8306	.6681	.0030
Small	Sinusoidal	5.0421	3.3860	.9110
Small	RoPE	6.7887	6.2249	3.0396

This demonstrates bounded positional and transport mechanics: the logical prompt exceeds one native operation, source-relative coordinates remain continuous, and no measured operation breaks its limit. It does not establish full functional equivalence to dense processing of all 192 tokens through every layer.

9 Related Work and Positioning

RoPE makes attention depend on relative displacement through position-dependent rotations [10]. Position Interpolation and LongRoPE extend dense-context operating ranges through rescaling and progressive extension [1, 4]. ALiBi and Transformer-XL provide other relative or distance-aware treatments [9, 2]. This paper is not another RoPE scaling recipe: it studies the ownership and rebinding of positions when K/V is encoded in chunks, stored externally, and retrieved sparsely.

Shortformer separates absolute positional information from reusable values [8]. Prompt Cache uses position-aware schemas for modular cache reuse, and CacheBlend selectively recomputes independently cached chunks that lack preceding context [5, 14]. MiniPIC similarly delays RoPE binding, while SemPIC learns a cache writer to address residual context dependence [7, 12]. These systems reinforce the distinction between position assignment and contextual hidden-state construction.

StreamingLLM preserves selected chronological decode states [11]. PRA instead routes among URI- or request-addressed chunks and must define both source identity and position semantics. Sparse attention and KV compression concern which state remains active or resident; the present question is whether a retained state still represents the coordinate and context its consumer assumes.

10 Limitations and Scope

The models are tiny and small controlled transformers, not pretrained production LLMs. The HotpotQA- and QASPER-derived tasks retain natural source text but replace open-ended answering with answer codes [13, 3]. WikiText-2 tests next-token behavior in bounded windows [6]. These choices isolate mechanisms but limit external validity.

Five paired seeds provide direction counts but weak inferential resolution: the smallest exact two-sided sign-flip result with five pairs is $p = 0.0625$. Formula-defined RoPE and sinusoidal coor-

dinates can be evaluated outside the training range, but mathematical availability is not evidence of learned extrapolation. Learned absolute positions remain bounded by their table.

Overlap approximates missing left context through additional encoding and does not recreate a fully dense history at every layer. The evidence oracle is a programmatic source-identity control, not a guarantee of optimal context composition. Deferred-RoPE timing uses an unfused prototype path on one older GPU and should not be projected to production serving.

This paper deliberately excludes pretrained Hugging Face adaptation, learned memory-use training, recursive multi-hop retrieval, multi-gist representation search, semantic-distance policies beyond the reported mechanism probes, and serving integration. Those extensions must not be read back into the present evidence.

11 Conclusion

Source-relative positional continuity is a general requirement for chunked and retrieved native K/V. A logical start offset repairs reset fragmentation exactly at layer zero for learned absolute, sinusoidal, and RoPE models, and improves deeper representation in every paired test. RoPE’s special value is not ownership of that offset idea; it is the explicit algebra that relates pre- and post-rotation keys and permits retrieval-time rebinding.

The residuals identify the rest of the problem. Overlap can reduce contextual fragmentation, but it costs extra encoding. Better representation geometry does not guarantee better language-model loss, retrieval, or answers. Routing and useful memory composition remain independent empirical questions.

That separation is the contribution. A universal result in which offsets improved every metric would have hidden the architecture of the failure. The measured dissociation shows where positional correctness ends and where semantic memory use begins.

A Detailed Task Results

Table 5: WikiText-2 at an 8:1 logical-to-native ratio. Deltas are intervention minus baseline; negative loss is better. Every row contains five paired seeds.

Tier	Position	Δ final-K	Δ offset loss	Better	Δ overlap loss	Better
Tiny	Learned absolute	-.4281	+.0200	0/5	-.0173	5/5
Tiny	Sinusoidal	-.1897	+.0040	1/5	-.0007	4/5
Tiny	RoPE	-.6292	-.0000	2/5	-.0048	5/5
Small	Learned absolute	-.4044	+.0292	0/5	-.0140	5/5
Small	Sinusoidal	-.1810	+.0707	0/5	-.0102	5/5
Small	RoPE	-.6314	-.0000	2/5	-.0148	5/5

Table 6: Split-5 routed answer-token loss. Better counts paired offset-versus-reset seeds.

Dataset	Tier	Position	Reset	Offset	Offset + overlap	Better
HotpotQA	Tiny	Absolute	.559	.482	.482	4/5
HotpotQA	Tiny	Sinusoidal	.359	.304	.358	3/5
HotpotQA	Tiny	RoPE	.948	1.056	1.080	2/5
HotpotQA	Small	Absolute	.245	.043	.068	3/5
HotpotQA	Small	Sinusoidal	.061	.017	.012	4/5
HotpotQA	Small	RoPE	.129	.170	.196	1/5
QASPER	Tiny	Absolute	.629	.309	.301	4/5
QASPER	Tiny	Sinusoidal	.375	.244	.244	4/5
QASPER	Tiny	RoPE	.683	.869	.871	0/5
QASPER	Small	Absolute	.460	.299	.298	4/5
QASPER	Small	Sinusoidal	.403	.003	.007	5/5
QASPER	Small	RoPE	.169	.324	.184	1/5

B RoPE Storage Timing

Table 7: Small-tier CUDA timing in ms, averaged over five seeded runs of 500 warm iterations.

Storage	Selected K/V	Gather	Reconstruct	RoPE	Attention	Warm
Post-position	32	.0312	.0000	.0000	.1343	.1842
Pre-position + deferred	32	.0337	.0191	.4021	.1343	.6511
Post-position	160	.0327	.0000	.0000	.1342	.1791
Pre-position + deferred	160	.0329	.0145	.3838	.1342	.6487

C Implementation and Reproducibility

The implementation separates the native operation limit from position capacity. Each cached layer records token K/V, source identity, logical start/end, and whether K is pre- or post-position. Contiguous token IDs are reconstructed as `logical_start + arange(length)`. Values are never rotated. Conversion copies the trained SelfAttention Q/K/V and output projections into PRA; with memory disabled, converted logits match the source model to test tolerance.

The final validation artifacts use commit `64f7cdf89b55`; generated cross-domain summaries use `052ae05`. The original decomposition and storage matrix use `d36f67752701`. Raw CSV/JSON, plots, expectations, and manifests are stored in the Paper 1.5 results directory. Tests cover large logical offsets, overlap and splitting, independent operation/position bounds, cache metadata, no-memory conversion parity, pre/post equivalence, deliberate rebinding, rollover, and CPU/GPU parity.

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